EMS-GCN: An End-to-End Mixhop Superpixel-Based Graph Convolutional Network for Hyperspectral Image Classification

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***Abstract*— The lack of labels is one of the major challenges in hyperspectral image (HSI) classification. Widely used Deep Learning (DL) models such as convolutional neural networks (CNNs) experience serious performance degradation when training samples are limited. In contrast, graph convolutional networks (GCNs) can simultaneously exploit the insufficient labeled data and massive unlabeled data of HSI in a semisupervised learning fashion. However, in order to reduce computational cost and mitigate noise, existing GCN-based classification methods usually perform superpixel segmentation as a preprocessing step and implement feature extraction as well as node classification on the predefined superpixel graph, where one superpixel might incorporate pixels with different labels. Moreover, the local spectral–spatial information within superpixels is generally ignored. To alleviate these two issues, we propose an end-to-end mixhop superpixel-based GCN (EMS-GCN) framework for HSI classification. Specifically, we first introduce the differentiable superpixel segmentation algorithm to map the pixel representations into a superpixel feature space, which allows refining the superpixel boundary with the training of the network. After that, a superpixel graph is constructed and fed into a novel mixhop superpixel-based GCN, where both the local information within superpixels and long-range information among superpixels are extracted, while the structure of the superpixel graph is updated at the same time. Finally, the enhanced superpixel representations are mapped back into a pixel feature space to conduct pixel-wise classification. Extensive experiments demonstrate the effectiveness of the proposed EMS-GCN method compared with other state-of-the-art methods.**

***Index Terms*— Graph convolutional network (GCN), hyperspectral image (HSI) classification, semisupervised learning, superpixel segmentation.**

I. Introduction

HYPERSPECTRAL images (HSIs) are the images with one continuous spectrum measured for each pixel, which contain abundant spatial and spectral information [1]. Hence, they can be used to identify and quantify various land covers and have been extensively applied in many areas, such as disease diagnosis, resource investigation, and military reconnaissance [2]–[4]. Among these applications, HSI classification is a common task and crucial procedure, aiming to predict the class label of every pixel in HSI data.

Generally, the pipeline of HSI classification consists of two steps, classifier design and feature extraction. Early-stage HSI classification methods focused on directly applying the machine learning classifiers to hyperspectral data, including the support vector machine (SVM), random forest (RF), multinomial logistic regression (MLR), and so on. Some typical representation-based classifiers such as sparse representation [5] and collaborative representation [6] have also been applied to HSI classification. However, these classifiers often consider the original spectral information of pixels in HSI while ignoring their surroundings. As a result, they are vulnerable to the spectral variability in raw HSI, and suffer from serious salt and pepper noise in the classification maps. In this case, some researchers devoted themselves to extracting more discriminative spectral or spatial features of HSI for classification. For instance, the principal component analysis (PCA) method is the most frequently used strategy to reduce redundancy and extract spectral information of HSI [3]. Handcrafted spatial features extracted by extended morphological profiles (EMPs) [7], morphological attribute profiles (APs) [8], and Gabor filterings [9] have also been exploited as input data for machine learning classifiers. With the enhanced spectral or spatial features, the intrinsic information of HSI could be revealed and the irrelevant information could be reduced. Nevertheless, these shallow classifiers with handcrafted features still fail to capture the underlying high-level semantic information of HSI and thus achieve unsatisfactory classification performance.

In recent years, Deep Learning (DL) technology was quite successful in tackling many tasks of computer vision, from image classification [10], [11] to semantic segmentation [12], [13]. Owing to their capacity to automatically extract complex and hierarchical features using multiple layered network architectures, many deep networks have been exploited for HSI classification. At the beginning, the stacked autoencoder (SAE) [14] and deep belief network (DBN) [15] were adopted as the deep network architectures for HSI classification. However, the input of these two models can only be 1-D vectors, which cannot exploit the original spatial structure of HSI. To alleviate this issue, convolutional neural

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networks (CNNs) with 2-D convolutions or 3-D convolutions have been widely applied to HSI classification, which can retain the original data structure and extract the spatial or spectral–spatial features of HSI. For example, Yao *et al.* [16] fed the cluster map that was the compressed representations of raw HSI into a 2-D CNN to capture spatial features for HSI classification. Hong *et al.* [17] captured the spectral–spatial information with a 2-D CNN from different modalities to enhance the performance of HSI classification. In [18], 2-D convolutions were combined with nonlinear morphological operators to construct a CNN model for spectral–spatial classification, which helps to generate semantically meaningful features. Based on ResNet, Paoletti *et al.* [19] have proposed a deeper residual 2-D CNN to learn deep-seated features from original HSI. Furthermore, emerging network architectures such as DenseNet [20] and GhostNet [21] founded on 2-D convolutions were also introduced for HSI classification. To better exploit the inherent 3-D structure of HSI, many researchers paid more attention to 3-D CNNs. In [22], a spectral–spatial residual network has been presented based on the 3-D convolutional layer. In addition, Hamida *et al.* [23] enabled a joint spectral and spatial information process through a 3-D CNN architecture. In addition to developing complex CNNs, several works focused on the construction of lightweight CNNs for HSI classification by modifying the standard convolution layer [24], [25]. However, these models may improve computational efficiency at a cost of accuracy. Meanwhile, the popular recurrent neural networks (RNNs) [26], [27], fully convolutional networks (FCNs) [28], [29], and generative adversarial networks (GANs) [30] also provide new insights into HSI classification. Moreover, as a hot topic in computer vision, attention mechanism has also been broadly investigated in HSI classification [31]–[33], which is usually combined with neural networks to further enhance features. In [34], the pure self-attention-based transformer was also applied for HSI classification by regarding the spectral signatures as sequence data. Despite the success of the above-mentioned DL methods, many studies have indicated that these deep models tend to suffer from serious performance degradation due to small number of samples for training, which limits their applicability [35], [36].

Fortunately, the graph convolutional networks (GCNs) [37] are competent to deal with the problem by simultaneously exploiting the labeled and unlabeled samples in the training stage, which can not only reduce the demand for manually labeled data but also model the latent relationships between the labeled and unlabeled data, thus getting a better understanding of HSI data. Several GCN-based HSI classification methods have been well devised by considering the whole HSI as the graph structure data with nodes and edges. Specifically, in [38], a spectral–spatial GCN was devised to extract features by simultaneously considering the spectral and spatial neighborhoods of each pixel. Mou *et al.* [39] further enlarged the spatial context of each pixel by learning a nonlocal graph structure. However, these methods treated each pixel in HSI as a graph node, which will inevitably incur huge computational cost. Therefore, in [40], a small-scale superpixel graph was first constructed and then updated through the multiscale graph convolution. Zeng *et al.* [41] also segmented the HSI into a set of superpixels and applied the cluster-GCN [42] algorithm on the superpixel graph. Since the superpixel-based GCN can only extract the long-range information, Liu *et al.* [43] combined the superpixel-based GCN with CNN to extract the superpixel-level long-range information and the pixel-level local information. Nevertheless, there are still two limitations of these superpixel-based GCN methods. On the one hand, because of the noise and spectral variability of original HSI data, pixels with different labels may be incorporated in one superpixel when performing superpixel segmentation as a preprocessing step. This may lead to misclassification in the classification map, especially for the classification boundary regions. On the other hand, pre-existing superpixel-based GCNs only allow for the long-range information propagation between adjacent superpixels, while ignoring the local spectral–spatial information within superpixels, which is particularly useful for the classification of HSI datasets with many small-scale and irregular objects.

In view of the aforementioned problems, this article proposes an end-to-end mixhop superpixel-based GCN (EMS-GCN) framework for HSI classification to refine the superpixel boundary as well as capture the local information within superpixels and the long-range information among superpixels simultaneously. First, the learnable CNN-based superpixel segmentation algorithm is introduced and modified to map the pixel features of HSI into a superpixel feature space, which is conducive to generating more precise superpixel boundary. Second, a superpixel graph is constructed with the superpixel representations and binary adjacency matrix. Third, we elaborately design a novel mixhop superpixel-based GCN with the combination of the mixhop graph convolutional layer and the channel attention module. The former can capture both the local and long-range superpixel representations as well as refine the superpixel graph structure. The latter helps to adaptively integrate local and long-range information and reconstruct more discriminative representation. Finally, we project the enhanced superpixel features back into a pixel feature space to conduct pixel-wise classification. Through the integration of learnable superpixel segmentation module, mixhop graph convolutional layer, and channel attention module, the EMS-GCN aims at well preserving object edges and adaptively exploiting the local and long-range information in different HSI scenes for better classification performance.

The main contributions of this article are as follows.

1) We propose an end-to-end superpixel-based GCN classification method by introducing a modified learning-based superpixel algorithm specially for HSI, which applies a new multiscale spectral–spatial CNN as feature extractor and an adaptive clustering distance. Therefore, more precise superpixel boundaries can be generated and detailed information within superpixels can be preserved to boost the classification performance.
2) We devise a novel mixhop superpixel-based GCN module to adaptively integrate the local and long-range superpixel representations as well as to refine the superpixel graph structure.
3) We conduct extensive experiments on three benchmark datasets to validate the performance of the proposed EMS-GCN method qualitatively and quantitatively.

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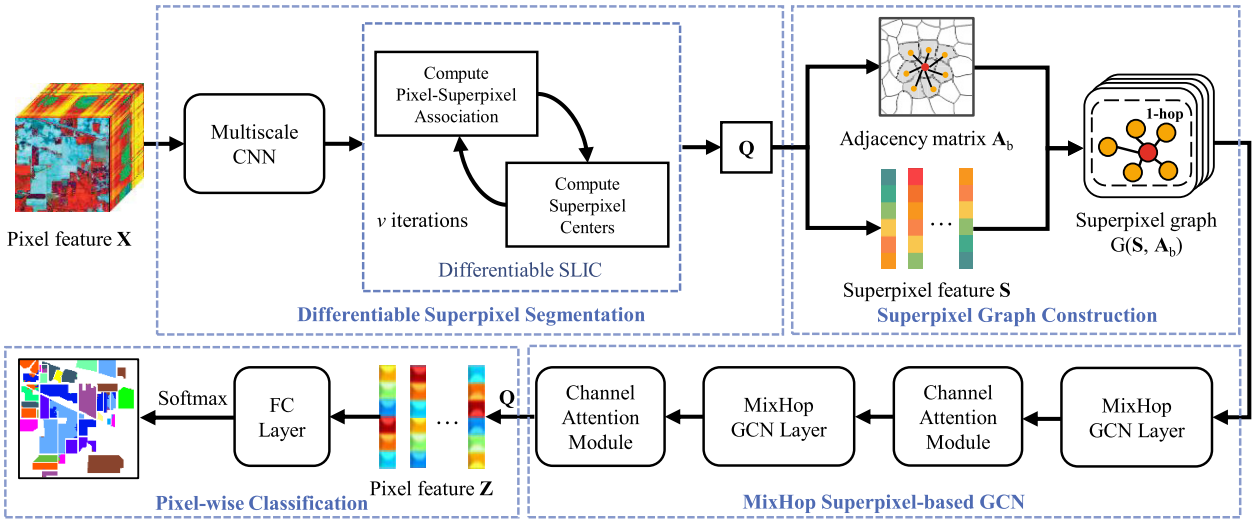

Fig. 1. Overall framework of EMS-GCN. The EMS-GCN framework consists of four modules: differentiable superpixel segmentation, superpixel graph construction, mixhop superpixel-based GCN, and pixel-wise classification.

Experimental results demonstrate the effectiveness and efficiency of our network.

The remaining of the article is organized as follows. Section II elaborates on the proposed EMS-GCN method with details. Section III exhibits the extensive experiments and analyses. Section IV presents the conclusions and hints at plausible future research work.

II. Proposed Method

A. Overall Framework

The architecture of the EMS-GCN method is shown in Fig. 1. Specifically, EMS-GCN consists of four main modules: 1) Differentiable Superpixel Segmentation; 2) Superpixel Graph Construction; 3) Mixhop Superpixel-based GCN; and 4) Pixel-wise Classification. The first module maps the pixel representations of HSI to a superpixel feature space in a differentiable way and the second module is used for superpixel graph construction. After that, the superpixel graph is fed into the third module for superpixel-level feature extraction together with graph structure update. Finally, the last module projects the enhanced superpixel features to pixel features and conducts pixel-wise classification. To optimize all of the parameters of EMS-GCN, the cross-entropy loss function is utilized to reduce the difference between the predicted labels and ground-truth of training samples. In addition, we employ the task-specific reconstruction loss function and compactness loss function to further supervise the learning process of the differentiable superpixel segmentation module, intending to learn task-specific superpixels as well as reduce the spatial variance of superpixels. Some key notations used throughout this article are summarized in Table I. The details of the four modules are described as follows.

TABLE I
Definitions of the Important Notations

Module	Notation	Definition
-	$\mathbf{X}$	Original HSI data
Differentiable Superpixel Segmentation	$\mathbf{I}^{xy}$	Positional pixel features
	$\mathbf{P}$, $\mathbf{P}'$	Flattened multiscale pixel features, concatenated pixel features
	$\mathbf{Q}$, $\hat{\mathbf{Q}}$	Original, column-normalized pixel-superpixel association matrices
Superpixel Graph Construction & MixHop Superpixel-based GCN	$\mathbf{H}$, $\mathbf{A}$, $\hat{\mathbf{A}}$	Node feature matrix, adjacency matrix, and normalized adjacency matrix for any graph
	$\mathbf{S}$, $\mathbf{A}_\mathrm{b}$, $\hat{\mathbf{A}}_\mathrm{b}$	Superpixel features, binary adjacency matrix, and normalized binary adjacency matrix for superpixel graph
	$\mathbf{S}_\mathrm{attn}$, $\mathbf{A}_\mathrm{w}$, $\hat{\mathbf{A}}_\mathrm{w}$	Attention enhanced superpixel features, weighted adjacency matrix, and normalized weighted adjacency matrix for superpixel graph
Pixel-wise Classification	$\tilde{\mathbf{Q}}$	Row-normalized pixel-superpixel association matrix
	$\mathbf{Q}'$	Hard-association matrix
	$\bar{\mathbf{I}}^{xy}$	Reconstructed positional features
	$\mathbf{Z}$	Reprojected pixel features

B. Differentiable Superpixel Segmentation

Taking every pixel in HSI as a graph node brings heavy computation burden and memory cost. As a result, many GCN-based HSI classification methods generally use superpixel segmentation as a preprocessing step to cluster a large number of HSI pixels into a few superpixels. However, the noise and spectral variability of the raw HSI impede the generation of convincing superpixels. Inspired by learning-based superpixel segmentation [44], we optimize the superpixel segmentation result by the modified CNN-based superpixel segmentation algorithm, which combines a multiscale CNN submodule with the differentiable simple linear iterative clustering (SLIC) submodule, as illustrated in Fig. 2.

Specifically, the original HSI data $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ are used as the input of CNN to extract D-dimensional multiscale pixel features. After that, we flatten these features into $\mathbf{P} \in \mathbb{R}^{N\times D}$, where H, W, B, N, and D represent the height, width, band number, total pixel number and dimension of hidden features,

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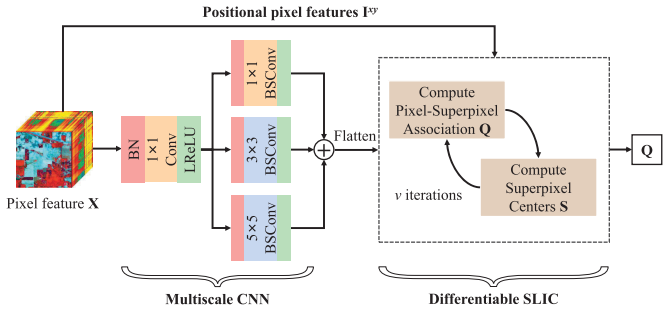

Fig. 2. Illustration of the differentiable superpixel segmentation module.

respectively. Subsequently, the flattened multiscale features $\mathbf{P}$ are concatenated with the two-dimensional positional pixel features $\mathbf{I}^{xy} \in \mathbb{R}^{N\times 2}$ to control the spatial compactness of superpixels, and the resulting pixel features $\mathbf{P}' \in \mathbb{R}^{N\times(D+2)}$ are fed into the differentiable SLIC submodule that alternatively updates the pixel-superpixel associations and superpixel centers for v iterations. The final output of the differentiable superpixel segmentation module is a pixel-superpixel association $\mathbf{Q} \in \mathbb{R}^{M\times N}$, which can transform pixel features to superpixel features by simple matrix multiplication, where M is the number of superpixels. In this article, M is determined by $M = H \times W/\lambda$, where λ represents the segmentation scale parameter.

The multiscale CNN submodule works as a learnable feature extractor to capture the discriminative and robust spectral–spatial information of HSI. Given that superpixel segmentation is on the basis of the low-level spectral and spatial features rather than the high-level semantic features, we devise a two-layer multiscale CNN based on convolution filters with different kernel sizes. Each convolutional block orderly includes a batch normalization (BN) layer, $k \times k$ convolution layer, and a leaky rectified linear units (LeakyReLU) activation function [45]. In addition, to avoid overfitting under limited training samples, the blueprint separable convolution (BSConv) [46] is applied to reduce the redundant parameters in standard convolution by combining a pointwise convolution and a depthwise convolution, which corresponds to a reversed depth-wise separable convolution (DSConv). Specifically, we use BSConv to substitute the original 3×3 convolution and 5×5 convolution.

The differentiable SLIC submodule can be considered as a soft k-means or expectation-maximization (EM) process [47], with $\mathbf{Q}$ and $\mathbf{S}$ alternately updated for v iterations. Concretely, superpixel features $\mathbf{S}$ are first initialized with the average pixel features in the corresponding regular grids. Afterward, with the pixel features $\mathbf{P}'$ and initial superpixel features $\mathbf{S}^{(0)}$, the soft association between the ith pixel and the jth superpixel at the tth iteration can be computed by

$$\mathbf{Q}_{ij}^{(t)} = \mathrm{e}^{-\frac{\left\|\mathbf{p}_i' - \mathbf{s}_j^{(t-1)}\right\|^2}{\beta^2}} \tag{1}$$

where $\mathbf{p}_i'$ is the feature vector of the ith pixel, $\mathbf{s}_j^{(t-1)}$ is the feature vector of the jth superpixel at the $(t-1)$th iteration, and β is a learnable parameter used to adaptively adjust the cluster distance for different datasets. We initially set it to 1 and update it with other network parameters in the training stage. It is noteworthy that $\mathbf{P}'$ rather than $\mathbf{P}$ is used to restrict the spatial variance of each superpixel.

Then we can map the pixel features into a new superpixel feature space by

$$\mathbf{S}^{(t)} = \hat{\mathbf{Q}}^{(t)^T}\mathbf{P}' \tag{2}$$

where $\hat{\mathbf{Q}}$ is the column-normalized $\mathbf{Q}$. By relaxing the definition of pixel-superpixel assignment coefficients from strictly 0 or 1, each entry in $\hat{\mathbf{Q}}$ can range from any value from 0 to 1, which will lead to weighted superpixel representations rather than average superpixel representations. The last two steps will be repeated for several iterations until the algorithm has converged. The interested reader can refer to [44], [48] for more details.

C. Superpixel Graph Construction

Here, we will introduce how to utilize the pixel-superpixel association $\mathbf{Q}$ to create the superpixel graph in detail. Let us consider the whole HSI as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the node sets and edge sets, respectively. In general, the $\mathcal{V}$ and $\mathcal{E}$ are depicted by the node feature matrix $\mathbf{H}$ and adjacency matrix $\mathbf{A}$ for ease of mathematical expression, where $\mathbf{h}_i$ represents the feature vector of node i, and $\mathbf{A}_{ij}$ represents the edge weight between node i and node j. In addition, we denote the diagonal degree matrix as $\mathbf{D}$ and the graph Laplacian matrix as $\mathbf{L}$.

On the one hand, the node feature matrix $\mathbf{H}$ of the superpixel graph is instantiated with the superpixel representations $\mathbf{S} \in \mathbb{R}^{M\times D}$, which can be calculated as follows:

$$\mathbf{H} = \mathbf{S} = \hat{\mathbf{Q}}^T\mathbf{P}. \tag{3}$$

On the other hand, according to the superpixel segmentation result, we first simply construct a binary adjacency matrix $\mathbf{A}_{\mathrm{b}} \in \mathbb{R}^{M\times M}$ to reflect the original superpixel graph structure, where $\mathbf{A}_{\mathrm{b}_{ij}}$ equals one if the ith node is adjacent with the jth node and zero otherwise. Note that two superpixels that share a common boundary are considered to be adjacent. In other words, for every superpixel node, we specify the value of one to its one-hop neighbors and zero to other superpixel nodes, which can be formulated as

$$\mathbf{A}_{ij} = \mathbf{A}_{\mathrm{b}_{ij}} = \begin{cases} 1, & \text{if } S_i \text{ and } S_j \text{ are adjacent} \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

However, it is unreasonable to assign the same edge weights (i.e., 1) for all adjacent node pairs. Therefore, we will update the superpixel graph structure via the similarity scores of learned superpixel representations in the subsequent mixhop superpixel-based GCN module, as detailed in Section II-D.

D. Mixhop Superpixel-Based GCN

By applying the convolution theorem, the convolution of a graph signal $\mathbf{s}$ can be formulated by first transferring the graph signal to the spectral domain and then applying a spectral filter g_θ parameterized by θ, that is

$$g_\theta * \mathbf{s} = \mathbf{U} g_\theta \mathbf{U}^T \mathbf{s} \tag{5}$$

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where $*$ represents the convolution operation, $\mathbf{U}$ denotes the eigenvector matrix of the Laplacian matrix $\mathbf{L}$ and can be computed by matrix decomposition, i.e., $\mathbf{L} = \mathbf{U}\Lambda\mathbf{U}^T$, where Λ is the diagonal matrix of eigenvalues.

Naive implementation requires high computational cost resulting from the explicit decomposition of the Laplacian matrix. To address this problem, Defferrard *et al.* [49] approximated the spectral filter g_θ concerning the Chebyshev polynomials up to Kth order. Recently, the formulation was further simplified by restricting $K = 1$, setting the largest eigenvalue of $\mathbf{L}$ to 2, and introducing a normalization trick to $\mathbf{A}$ [37]. With the three aforementioned simplifications, the graph convolution turns into

$$\mathbf{H}^{(l+1)} = \phi\left(\hat{\mathbf{A}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{6}$$

where $\mathbf{H}^{(l)}$ is the node feature matrix of layer l and $\mathbf{H}^{(l+1)}$ is the node feature matrix of layer $(l+1)$, $\mathbf{W}^{(l)}$ is the weight matrix of layer l, and ϕ represents an activation function. $\hat{\mathbf{A}}$ denotes the normalized adjacency matrix with self-connections and can be computed by

$$\hat{\mathbf{A}} = \tilde{\mathbf{D}}\tilde{\mathbf{A}}\tilde{\mathbf{D}} \tag{7}$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\tilde{\mathbf{D}} = \sum_j \tilde{\mathbf{A}}_{ij}$.

Equation (6) lays the foundation of conducting convolution on graph structure data, as adopted in some related works [39]–[43]. In terms of the convolution of superpixel graph, (6) can be rewritten as

$$\mathbf{S}^{(l+1)} = \phi\left(\hat{\mathbf{A}}_{\mathrm{b}}\mathbf{S}^{(l)}\mathbf{W}^{(l)}\right) \tag{8}$$

where $\hat{\mathbf{A}}_{\mathrm{b}}$ represents the normalized binary adjacency matrix with self-connections. It is observed that the vanilla graph convolutional layer can only aggregate the long-range information among the one-hop neighbors of superpixels. In contrast, we are interested in both the local and long-range information in HSI data, which separately corresponds to the features within superpixels and among superpixels. Therefore, motivated by [50], we devise a mixhop graph convolutional layer to focus on zero-hop and one-hop neighborhood information simultaneously, which is defined as

$$\mathbf{S}^{(l+1)} = \|_{m\in\{0,1\}}\phi\left(\hat{\mathbf{A}}_{\mathrm{w}}^{m}\mathbf{S}^{(l)}\mathbf{W}_m^{(l)}\right) \tag{9}$$

where $\hat{\mathbf{A}}_{\mathrm{w}}^{m}$ represents the normalized weighted adjacency matrix with self-connections $\hat{\mathbf{A}}_{\mathrm{w}}$ multiplied by itself m times and $\|$ represents the column-wise concatenation. $\hat{\mathbf{A}}_{\mathrm{w}}^{0}$ denotes the identity matrix $\mathbf{I}$ and $\hat{\mathbf{A}}_{\mathrm{w}}^{1}$ denotes $\hat{\mathbf{A}}_{\mathrm{w}}$ itself. In addition, there are two parameter matrices $\mathbf{W}_m^{(l)} \in \mathbb{R}^{M\times F_m^{(l)}}$, where $F_m^{(l)}$ is the feature dimension of the mth parameter matrix at the lth layer. We set the two matrices to the same dimension and equal to half of the feature dimension in vanilla graph convolutional layer by default, that is $F_0^{(l)} = F_1^{(l)} = F^{(l)}/2$, where $F^{(l)}$ represents the feature dimension in vanilla graph convolutional layer at the lth layer.

The comparison of the vanilla graph convolutional layer and mixhop graph convolutional layer is shown in Fig. 3, and there are two main differences between them. For one thing, the mixhop graph convolutional layer can capture both the local and long-range information of superpixels, while the vanilla graph convolutional layer fails to extract the local features within superpixels. For another, on the basis of the binary adjacency matrix $\mathbf{A}_{\mathrm{b}}$, a weighted adjacency matrix $\mathbf{A}_{\mathrm{w}}$ is adaptively computed in the mixhop graph convolutional layer. In detail, we specify different edge weights for the binary adjacency matrix $\mathbf{A}_{\mathrm{b}}$ via embedded dot product of a pair of node features, which can be formulated as

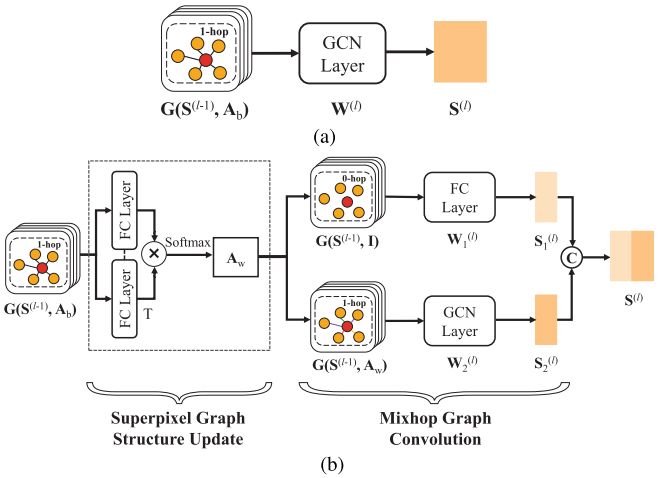

Fig. 3. Comparison of the vanilla graph convolutional layer and mixhop graph convolutional layer. (a) Vanilla graph convolutional layer. (b) Mixhop graph convolutional layer.

$$\mathbf{A}_{\mathrm{w}_{ij}} = \begin{cases} \mathrm{Sigmoid}\left(\left(\mathbf{W}_\varphi \mathbf{s}_i\right)\left(\mathbf{W}_\varphi \mathbf{s}_j\right)\right), & \text{if } \mathbf{A}_{b_{ij}} = 1 \\ \alpha, & \text{otherwise} \end{cases} \tag{10}$$

where $\mathbf{W}_\varphi \mathbf{s}_i$ and $\mathbf{W}_\varphi \mathbf{s}_j$ are two embeddings with shared weights, α is the mask values before activation (a large negative real number like -10^9) and $\mathbf{A}_{\mathrm{w}_{ij}}$ represents the learned edge weight between node i and node j. The sigmoid function is used to prevent numerical instability.

At last, we add self-connections (identity matrix $\mathbf{I}$) to $\mathbf{A}_{\mathrm{w}}$ and then apply the softmax activation function with the aim of normalization, namely

$$\hat{\mathbf{A}}_{\mathrm{w}} = \mathrm{Softmax}(\mathbf{A}_{\mathrm{w}} + \mathbf{I}). \tag{11}$$

Note that the softmax function instead of the degree matrix is applied for normalization to accelerate the training speed. The mixhop graph convolutional layer needs little additional computational complexity and memory consumption, and can be a plug-and-play replacement for vanilla graph convolutional layer on superpixel graph for better classification performance.

Since the effect of local and long-range information is different in various HSI scenes, it is necessary to determine what kind of spatial context is more valuable to ensure accurate classification. For this reason, we further design a channel attention module inserted after the mixhop graph convolutional layer, which can adaptively learn the weight of each channel for and reconstruct more discriminative feature representations. For the channel attention module shown in Fig. 4, we first generate a diagonal matrix $\Lambda(\mathbf{S}) \in \mathbb{R}^{F\times F}$ that represents the significance of each channel, and then multiply the original features $\mathbf{S}$ by $\Lambda(\mathbf{S})$ in order to enhance feature

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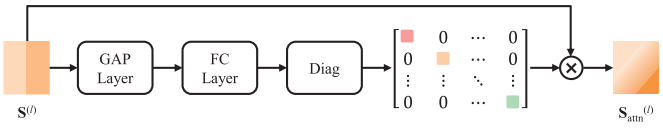

Fig. 4. Illustration of the channel attention module.

representations. The overall computation can be formulated as

$$\mathbf{S}_{\text{attn}} = \mathbf{S}\Lambda(\mathbf{S}). \tag{12}$$

Concretely, $\Lambda(\mathbf{S})$ is computed in a similar manner to the channel attention devised in [51] and we implement $\Lambda(\mathbf{S})$ as

$$\Lambda(\mathbf{S}) = \text{Diag}\big(\sigma\left(\mathbf{W}_\rho(\text{GAP}(\mathbf{S}))\right)\big) \tag{13}$$

where $\mathbf{S} \in \mathbb{R}^{M\times F}$ is the output feature matrix of a mixhop graph convolutional layer, GAP represents the global average pooling operation, $\mathbf{W}_\rho$ is the trainable matrix of fully connected layer, σ denotes the sigmoid function, and Diag stands for diagonalization. Note that the layer l is omitted in (12) and (13) for the sake of simplicity.

Consequently, we adopt two mixhop graph convolutional layers with the channel attention module in the proposed mixhop GCN to capture semantic information while preserving detailed information, and adaptively increasing the discrimination of superpixel representations.

E. Pixel-Wise Classification

With the output superpixel features $\mathbf{S}_{\text{attn}}^{(\text{D})}$ extracted by mixhop GCN and the row-normalized $\mathbf{Q}$ (i.e., $\tilde{\mathbf{Q}}$), we are able to map the superpixel features into a pixel feature space to conduct pixel-wise classification, that is

$$\mathbf{Z} = \tilde{\mathbf{Q}}\mathbf{S}_{\text{attn}}^{(\text{D})} \tag{14}$$

where D is the depth of the mixhop GCN (D equals 2 by default), and $\mathbf{Z}$ is the reprojected pixel representations. Note that, as each entry in $\tilde{\mathbf{Q}}$ can range from any value from 0 to 1, the reprojected pixels in one superpixel will have diverse, rather than identical feature representations, which is helpful for preserving the detailed information within superpixels. In the end, we use the softmax classifier including a fully connected layer and a softmax activation function to predict class probabilities of the reprojected pixel features.

For the HSI classification task, the commonly used cross-entropy loss function is applied to reduce the difference between the network predicted labels and ground truth of the training samples, that is

$$L_{\text{CE}}(\hat{\mathbf{Y}}, \mathbf{Y}) = -\frac{1}{T}\sum_{t=1}^{T}\sum_{c=1}^{C}\mathbf{Y}_c^t \log\left(\hat{\mathbf{Y}}_c^t\right) \tag{15}$$

where $\mathbf{Y}$ denotes the ground-truth and $\hat{\mathbf{Y}}$ denotes the predict labels, C represents the number of categories, and T is the total number of training samples.

Moreover, aiming to generate superpixels that stick to object boundary and have lower spatial variance, we combine a task-specific reconstruction loss function L_{recon} with a compactness loss function L_{cpt} to further supervise the differentiable superpixel segmentation module [44], which are, respectively, given by

$$L_{\text{recon}}(\mathbf{Q}, \mathbf{Y}) = L_{\text{CE}}\left(\tilde{\mathbf{Q}}\hat{\mathbf{Q}}^T\mathbf{Y}, \mathbf{Y}\right) \tag{16}$$

$$L_{\text{cpt}}\left(\mathbf{Q}, \mathbf{I}^{xy}\right) = \left\|\bar{\mathbf{I}}^{xy} - \mathbf{I}^{xy}\right\|_2 \tag{17}$$

where $\tilde{\mathbf{Q}}\hat{\mathbf{Q}}^T\mathbf{Y}$ represents the reconstructed semantic labels and $\bar{\mathbf{I}}^{xy}$ denotes the reconstructed positional pixel features, which can be computed by

$$\begin{aligned}\mathbf{S}^{xy} &= \hat{\mathbf{Q}}^T\mathbf{I}^{xy}\\ \bar{\mathbf{I}}_i^{xy} &= \mathbf{S}_j^{xy}\big|\mathbf{Q}_i' = j\end{aligned} \tag{18}$$

where the hard-association matrix $\mathbf{Q}'$ is determined as follows:

$$\mathbf{Q}_i' = \underset{j\in\{1,\ldots,M\}}{\arg\max}\ \mathbf{Q}_{ij}. \tag{19}$$

Hence, the joint loss function of the proposed EMS-GCN framework is written as

$$L = L_{\text{CE}}(\hat{\mathbf{Y}}, \mathbf{Y}) + w_1 L_{\text{recon}}(\mathbf{Q}, \mathbf{Y}) + w_2 L_{\text{cpt}}\left(\mathbf{Q}, \mathbf{I}^{xy}\right) \tag{20}$$

where w_1 and w_2 are two weight coefficients. Note that our model is trained in an end-to-end manner and the parameters in the model are optimized by backpropagation and full-batch stochastic gradient descent algorithm.

III. Experiments

A. Data Description

We evaluate the performance of the EMS-GCN method on three widely used HSI datasets, including Indian Pines (IP), Pavia University (PU), and Houston University (HU). In the following, each dataset is described in detail.

1) Indian Pines (IP): The IP dataset was obtained by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over Northwestern Indiana, United States (US), which is composed of 145 × 145 pixels with 20 m per pixel resolution and has 200 spectral bands after removing 20 water absorption channels. The image is divided into 16 classes of interest. The information about all 16 classes, as well as the amounts of training and test data for different classes, are detailed in Table II. And the false color image and ground truth map of the IP dataset are exhibited in Fig. 5.

2) Pavia University (PU): The PU dataset was collected by the reflective optics system imaging spectrometer (ROSIS) sensor over the city of Pavia, Italy. This image has 610 × 340 pixels and a high spatial resolution (1.3 m per pixel resolution). After removing 12 noisy bands, 103 spectral bands are used for analysis. The ground truth comprises nine categories. Table III details the category information and the number of labeled and unlabeled pixels for each class. Fig. 6 shows the false color image and ground truth map of the PU dataset.

3) Houston University (HU): The Compact Airborne Spectrographic Imager (CASI) sensor collected data at 2.5 m per pixel resolution over the University of Houston campus and the neighboring urban area, US, which comprises 1905 × 349 pixels and 144 spectral channels. 15 classes representing different land cover types are defined in the dataset. The dataset

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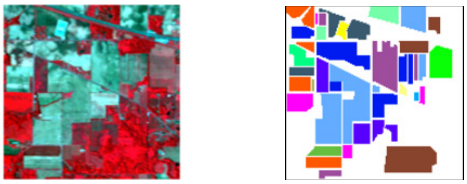
TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS OF THE IP DATASET

No.	Color	Class name	Training	Test
1		Alfalfa	30	16
2		Corn-notill	30	1398
3		Corn-mintill	30	800
4		Corn	30	207
5		Grass-pasture	30	453
6		Grass-trees	30	700
7		Grass-pasture-mowed	15	13
8		Hay-windrowed	30	448
9		Oats	15	5
10		Soybean-notill	30	942
11		Soybean-mintill	30	2425
12		Soybean-clean	30	563
13		Wheat	30	175
14		Woods	30	1235
15		Buildings-Grass-Trees-Drives	30	356
16		Stone-Steel-Towers	30	63
		Total	450	9799



(a) (b)

Fig. 5. IP dataset. (a) False color image. (b) Ground truth map.

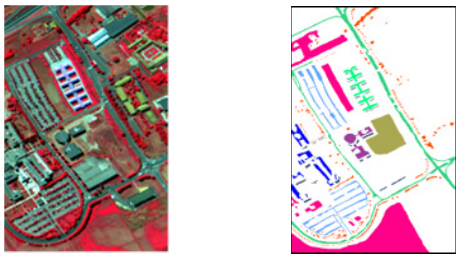
TABLE III
NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS OF THE PU DATASET

No.	Color	Class name	Training	Test
1		Asphalt	30	6601
2		Meadows	30	18619
3		Gravel	30	2069
4		Trees	30	3034
5		Painted metal sheets	30	1315
6		Bare Soil	30	4999
7		Bitumen	30	1300
8		Self-Blocking Bricks	30	3652
9		Shadows	30	917
		Total	270	42506

was used in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest, its complete category information and the number of training and test pixels for each class are listed in Table IV. The false color image and ground truth map of the HU dataset are displayed in Fig. 7.

TABLE IV
NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS OF THE HU DATASET

No.	Color	Class name	Training	Test
1		Grass Healthy	198	1053
2		Grass Stressed	190	1064
3		Grass Synthetic	192	505
4		Tree	188	1056
5		Soil	186	1056
6		Water	182	143
7		Residential	196	1072
8		Commercial	191	1053
9		Road	193	1059
10		Highway	191	1036
11		Railway	181	1054
12		Parking Lot1	192	1041
13		Parking Lot2	184	285
14		Tennis Court	181	247
15		Running Track	187	473
		Total	2832	12197



(a) (b)

Fig. 6. PU dataset. (a) False color image. (b) Ground truth map.

B. Experiment Setup

1) Implementation Details: In our experiments, we set the number of epochs as 600, 300, and 300 for the IP dataset, PU dataset, and HU dataset, respectively. Other configurations of our proposed EMS-GCN are identical for all datasets. Specifically, we adopt the Adam optimizer with a learning rate equal to 0.0005 to train our network and remain it unchanged during the whole process. The number of iteration υ is set to 5 for the differentiable SLIC submodule. As we consider the whole HSI as a superpixel graph, the batch size can only be set to 1. In addition, we set the weight coefficients w_1 and w_2 to 0.1 and 0.001, respectively. And we choose the LeakyReLU as the activation function of mixhop GCN. As to the hardware environment, our experiments have been conducted on an NVIDIA server with a Titan RTX 3090 GPU. On the other hand, the software environment consists of Python-3.8.5 and PyTorch-1.1.0. The other configurations of the EMS-GCN method are detailed in Table V. In terms of the training and test sets, we adopt two sampling strategies to provide a more comprehensive evaluation on the effectiveness of the proposed EMS-GCN method. Specifically, for the IP and PU datasets, we randomly select 30 samples from labeled pixels in each class for training (15 samples for the 7th and 9th classes in the IP dataset), the other pixels are used as the test samples. For

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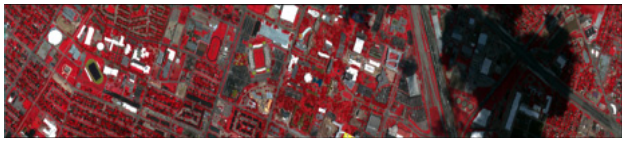

(a)

(b)

Fig. 7. HU dataset. (a) False color image. (b) Ground truth map.

TABLE V
CONFIGURATIONS OF EMS-GCN FOR ALL DATASETS (I.E., IP, PU, AND HU DATASETS)

Module	Layer	Configuration
Differentiable Superpixel Segmentation	conv 1-1	1×1@128
	conv 2-1	1×1@128
	bsconv 2-2	3×3@128
	bsconv 2-3	5×5@128
Superpixel Graph Construction	-	-
MixHop Superpixel-based GCN	fc 1-1	@256
	fc 1-2	@64
	graph conv 1-1	@64
	fc 1-3	@128
	fc 2-1	@256
	fc 2-2	@32
	graph conv 2-1	@32
	fc 2-3	@64
Pixel-wise Classification	fc 1-1	@#Classes

the HU dataset, we use the spatially disjoint training-test sets provided by the 2013 IEEE GRSS Data Fusion Contest. Note that we do not use validation set for hyperparameter selection. In addition, each experiment is repeated five times. The overall accuracy (OA), average accuracy (AA), kappa coefficient, and per-class accuracy (CA) are used as the evaluation criteria.

2) Comparison Methods: To evaluate the classification performance of the EMS-GCN method, ten HSI classification methods are selected for comparison, including three traditional methods, i.e., SVM, EMP-SVM [52], and Superpixel Graph Learning (SGL) [53] (SGL), four CNN-based methods, i.e., HybridSN [54] (Hybrid Spectral Network), pResNet [19] (Deep Pyramidal ResNet), A^2S^2K-ResNet [55] (Adaptive Spectral–Spatial Kernel ResNet), and Multiscale Residual Network (MSRN) [56] (MSRN), as well as three GCN-based methods, i.e., FuNet [57] (Fusion Network), Multi-scale Dynamic Graph Convolutional Network (MDGCN) [40] (MDGCN) and Convolutional Neural Network Enhanced Graph Convolutional Network (CEGCN) [43] (CEGCN).

For the SVM, we use five-fold cross validation to search for the optimal hyperparameters γ (kernel coefficient) in the range of {0.1, 0.01, 0.001} and C (penalization coefficient) in the range of {1, 10, 100, 1000}, and we choose the radial basis function (RBF) kernel by default. For the EMP-SVM, we first conduct several openings and closings operations (i.e., 7 for IP and PU datasets; 13 for HU dataset) on the first four principal components of the HSI data to extract structural features, where the disk shape starts with one-pixel size and increases one pixel each step. After that, SVM with grid search is used for classification. For the CNN-based and GCN-based comparative methods, we adopt the default values in open source codes or the hyperparameters recommended in original articles.

C. Classification Results

In this section, three datasets are used for the quantitative and qualitative evaluation of various classification methods.

1) Quantitative Evaluation: The mean and the standard deviation of each index (i.e., OA, AA, kappa coefficient, and CA) obtained by various classification methods on three datasets are reported in Tables VI–VIII, in which the bold number denotes the highest value and the underline number represents the second highest value. For traditional classification approaches, EMP-SVM outperforms SVM for the reason that it can fully exploit the abundant spatial information together with the spectral information of HSI data. By combining superpixels with a traditional graphical classifier, SGL exhibits a significant superiority over the other two traditional methods, and achieves higher performance than the CNN-based classification methods and the pixel-based GCN (i.e., FuNet) method, which demonstrates the benefit of the superpixel graph-based classification framework. Specifically, the superpixel segmentation technique is conducive to generating discriminative superpixel graph representations, and the graph-based classifier effectively captures the complex long-range dependencies among superpixel nodes based on the flexible graph structure. Compared with SGL, the superpixel-based GCN (i.e., MDGCN, CEGCN, and EMS-GCN) methods can automatically learn the high-level representations of superpixels rather than extract handcrafted superpixel features, which generally leads to higher classification accuracy.

In terms of the superpixel-based GCN models, CEGCN provides higher OA, AA, and kappa coefficients than MDGCN, mainly because it introduces a pixel-based CNN branch in addition to the superpixel-based GCN branch to preserve the local detailed information. Furthermore, without the help of the extra CNN branch, the proposed EMS-GCN still obtains the highest accuracy among all comparative methods in regard to OA and Kappa coefficient on three datasets. Specifically, as compared to the second best method, our proposed EMS-GCN increases the classification accuracy by 1.17% of OA, 0.29% of AA, and 1.33% of Kappa coefficient on the IP dataset, by 0.63% of OA, 0.20% of AA, and 0.83% of Kappa coefficient on the PU dataset, and by 0.13% of OA and 0.14% of Kappa coefficient on the HU dataset, which demonstrates the effectiveness of the EMS-GCN method for HSI classification.

2) Qualitative Evaluation: For the purpose of visual comparison, the classification maps of comparative methods are

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TABLE VI
ACCURACY COMPARISONS ON THE IP DATASET

ID	Traditional method			CNN-based method				GCN-based method			
	SVM	EMP-SVM	SGL	HybridSN	pResNet	A^2S^2K-ResNet	MSRN	FuNet	MDGCN	CEGCN	EMS-GCN
1	93.75±6.85	97.5±3.06	**100±0**	**100±0**	98.75±2.8	88.07±8.61	53.26±21.86	95.83±3.61	**100±0**	**100±0**	97.5±3.06
2	49.54±5.43	63.39±8.01	92.08±6.54	74.36±4.9	80.14±4.12	92.27±3.66	91.78±2.32	58.35±2.47	78.61±0.33	83.32±1.81	**92.8±1.94**
3	53.18±1.48	81.65±8.22	90.88±3.38	86.63±2.93	87.97±5.67	89.38±3.25	93.31±4.66	58.71±6.45	96.18±0.85	**97.94±1.5**	96.5±1.7
4	70.63±12.72	95.17±2.87	96.78±1.78	98.36±1.58	97.59±2.9	89.17±4.83	95.61±3.15	80.68±4.43	90.92±3.44	**99.88±0.21**	98.94±0.99
5	89.01±2	89.85±3.02	96.84±1.46	92.05±3.39	97.13±1.85	**98.39±0.82**	95.36±1.43	88.89±0.34	90.2±0.3	93.43±1.65	94.53±3.15
6	90.77±2.76	92.03±0.99	**99.57±0**	97.66±2.32	95.57±2.66	98.77±0.29	93.8±1.66	95.71±2.48	97.94±0.08	99.46±0.31	98.43±0.81
7	89.23±7.85	96.92±3.77	**100±0**	**100±0**	**100±0**	55.2±3.83	57.29±31.7	87.18±4.44	**100±0**	**100±0**	98.46±3.08
8	87.72±3.2	99.46±0.48	**100±0**	98.93±1.07	99.33±1.16	**100±0**	95.03±2.93	94.72±4.39	99.42±0.89	**100±0**	**100±0**
9	72±24	**100±0**	**100±0**	**100±0**	**100±0**	32.82±2.6	37.98±18.87	**100±0**	**100±0**	**100±0**	**100±0**
10	62.14±6.05	77.83±4.79	**92.85±1.08**	87.49±5.44	87.33±5.63	78.66±5.38	85.96±5.43	76.72±5.35	91.97±0.79	90.9±0.71	91.4±4.6
11	54.14±4.72	75.98±2.43	91.64±2.2	74.73±3.49	80.99±1.52	**96.27±0.6**	95.22±2.49	64.04±2.22	95.83±1.19	92.85±1.21	94.24±2.18
12	54.96±8.72	87.5±6.29	96.74±1.39	86.29±2.67	84.05±8.15	91.88±3.9	87.84±9.83	68.56±2.15	88.7±3.15	**98.67±0.2**	97.51±0.88
13	95.31±1.67	99.2±0.46	99.43±0	**100±0**	**100±0**	94.26±1.88	85.32±11.99	99.05±0.33	97.71±0	**100±0**	99.09±0.58
14	81.2±4.23	96.41±3.28	98.3±2.4	91.08±3.99	95.48±2.26	98.3±0.96	97.48±1.65	86.1±0.51	99.35±0.4	99.31±0.18	**99.94±0.13**
15	54.72±7.29	90.23±1.75	99.44±0.4	98.37±1.43	94.61±4.56	91.5±1.41	87.42±5.6	72±2.61	99.55±0.15	96.28±1.06	**99.83±0.34**
16	94.92±3.94	97.46±1.91	**100±0**	98.73±1.56	99.68±0.71	82.44±7.68	65.45±9.06	**100±0**	**100±0**	**100±0**	**100±0**
OA	64.88±0.89	82.47±1.59	94.70±1.14	85.20±1.85	88.05±1.02	92.77±1.06	91.91±0.45	73.46±0.19	93.23±0.33	94.16±0.42	**95.87±0.74**
AA	74.58±6.43	90.04±3.21	97.16±3.28	92.79±0.72	93.66±0.61	86.09±0.76	82.38±1.32	82.91±0.29	95.40±0.25	97.00±0.21	**97.45±0.47**
Kappa	60.31±0.92	80.07±1.81	93.94±1.30	83.28±2.02	86.40±1.15	91.74±1.20	90.77±0.51	69.92±0.31	92.25±0.37	93.32±0.48	**95.27±0.85**

TABLE VII
ACCURACY COMPARISONS ON THE PU DATASET

ID	Traditional method			CNN-based method				GCN-based method			
	SVM	EMP-SVM	SGL	HybridSN	pResNet	A^2S^2K-ResNet	MSRN	FuNet	MDGCN	CEGCN	EMS-GCN
1	73.85±3.59	93.4±0.8	90.82±2.35	91.33±3.36	92.71±1.34	92.48±1.94	98.38±1.08	79.1±4.17	93.55±0.37	**98.44±0.64**	98.26±1.14
2	76.71±5.87	82.7±7.38	99.13±0.6	93.65±2.62	95.25±3.06	98.28±0.42	**99.65±0.27**	92.55±3.06	99.25±0.23	96.38±0.15	97.66±1.26
3	75.99±4.12	88.62±3.68	99.17±0.46	87.91±4.54	85.8±9.1	69.53±1.59	97.4±1.6	74.17±3.53	92.03±0.24	99.79±0.19	**99.95±0.05**
4	89.55±2.49	95.81±2.25	89.02±2.58	93.48±2.7	96.01±1.9	90.6±8.22	90.91±4.28	95.06±2.95	83.78±1.55	96.35±0.73	**97.35±0.18**
5	99.21±0.1	99.65±0.04	99.29±1.21	**100±0**	99.55±0.52	99.15±0.7	96.99±2.48	**100±0**	99.47±0.09	**100±0**	99.81±0.19
6	82.23±7.31	80.25±6.65	**100±0**	86.68±5.8	90.78±4.14	80.47±5.33	92.46±6.58	85.18±1.01	95.26±0.50	99.89±0.15	**100±0**
7	89.69±4.06	95.22±0.71	99.66±0.25	99.05±1.32	91.42±5.98	72.56±4.5	97.73±2.25	87.41±2.17	98.92±1.04	**99.98±0.03**	99.96±0.04
8	80.59±5.68	88.87±4.9	92.86±2.05	78.42±9.29	93.48±5.2	68.38±4.94	92.06±2.91	79.51±5.16	94.99±1.33	99.46±0.49	**99.77±0.04**
9	99.85±0.09	99.94±0.05	98.58±0.98	98.78±0.47	99.59±0.47	93.22±1.7	91.39±3.66	**99.96±0.06**	81.03±0.49	99.89±0.08	99.24±0.65
OA	79.72±1.78	87.11±2.50	96.69±0.43	91.34±1.52	93.88±1.78	88.44±1.31	96.69±1.02	87.99±1.78	95.68±0.22	97.84±0.09	**98.47±0.73**
AA	85.3±3.70	91.61±2.94	96.5±4.08	92.14±1.62	93.84±1.47	84.96±1.92	95.22±0.69	88.11±1.16	93.15±0.28	98.91±0.03	**99.11±0.33**
Kappa	74.1±1.98	83.33±2.98	95.61±0.58	88.58±2.01	91.93±2.29	84.96±1.61	95.63±1.32	84.21±2.27	94.25±0.29	97.15±0.13	**97.98±0.96**

displayed in Figs. 8–10, respectively. In general, the spectral classifier SVM suffers from serious salt and pepper noise in the classification maps and EMP-SVM alleviates the problem by manually extracting spatial features. SGL can generate smoother classification maps with the regularization using superpixels. As expected, the CNN-based classification methods and GCN-based classification methods generally obtain smooth classification maps. Compared with other models, SGL and the superpixel-based GCN models are able to preserve more edge details in the classification maps, mainly due to the adaptive superpixel boundary and flexible graph propagation. For instance, fewer misclassifications in boundary regions can be observed for the second class ("Corn-notill") on the IP dataset, the second class ("Meadows") on the PU dataset, and the ninth class ("Road") on the HU dataset. Moreover, the classification maps produced by the EMS-GCN method show more consistent agreement with ground truth since it can not only suppress the misclassification and noisy scattered points in classification maps but also preserve the detailed information well.

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TABLE VIII
ACCURACY COMPARISONS ON THE HU DATASET

ID	Traditional method			CNN-based method				GCN-based method			
	SVM	EMP-SVM	SGL	HybridSN	pResNet	A^2S^2K-ResNet	MSRN	FuNet	MDGCN	CEGCN	EMS-GCN
1	82.62	81.01	78.63±0	82.68±0.74	81.98±0.82	98.5±1.96	**99.73±0.33**	82.94±0.2	81.23±0.55	82.75±0.3	82.89±0.95
2	83.93	84.12	81.02±1.42	84.79±0.38	84.32±0.76	96.11±4.8	**97.32±1.84**	84.37±0.57	84.02±0.85	84.42±0.53	84.98±0.04
3	99.8	99.6	**100±0**	96.44±1.42	92.59±2.09	99.73±0.26	96.88±6.98	83.5±1.6	99.21±0	**100±0**	99.96±0.08
4	98.49	90.25	67.14±0.59	81.61±3.97	91.8±1.47	80.55±11.72	**99.19±1.64**	91.89±1.04	82.86±1.73	98.28±2.94	91.74±1.35
5	98.96	99.53	99.65±0.44	99.92±0.11	99.45±0.55	93.52±2.51	99.76±0.31	99.53±0.58	**100±0**	**100±0**	99.62±0.35
6	98.6	99.3	95.8±0	**99.58±0.56**	95.52±4.44	50.98±37.03	56.8±36.86	97.44±1.46	95.8±0	96.78±1.37	96.36±1.84
7	75.56	**91.7**	70.8±3.09	68.06±5.06	85.86±2.3	90.27±5.29	79.73±14.37	78.98±3.8	70.09±0.97	85.76±2.25	83.84±2.25
8	61.35	41.98	68.28±2.21	**86.78±5.38**	73.18±3	73.91±15.03	71.08±12.08	67.58±3	74.93±1.98	75.73±4	85.22±6.42
9	66.1	78.47	74.66±0.91	74.58±9.03	75.22±2.8	**90.29±7.59**	90.24±4.22	81.27±3.65	75.98±1.44	88.18±1.2	**90.29±1.68**
10	61.97	64.29	76.58±5.83	53.57±8.6	60.56±2.34	75.57±19.48	**82.02±8.94**	53.15±2.92	67.7±0.75	68.13±0.38	65.23±4.33
11	76.85	93.83	**96.9±0.05**	77.31±10.94	71.08±2.98	81.01±2.96	81.49±16.66	64.9±5.45	75.71±1.4	90.15±5.29	92.28±6.33
12	67.72	59.85	93.4±1.87	85.48±6.89	90.14±3.21	71.76±12.33	94.12±4.62	81.78±2.11	74.61±0.39	97.46±1.25	**97.81±2.63**
13	64.91	70.53	74.62±1.42	80.49±8.2	82.04±2.85	**90.95±5.53**	83.48±3.7	88.65±1.13	48.42±0	84.07±3.5	78.95±5.29
14	99.6	**100**	98.79±0	99.92±0.16	87.78±6.04	96.35±5.16	99.6±0.69	92.31±1.86	**100±0**	**100±0**	99.68±0.65
15	97.46	**100**	**100±0**	99.79±0.42	75.9±9.13	99.27±0.28	94.94±2.93	95.98±3	99.3±0.85	99.79±0.27	99.96±0.08
OA	79.51	80.77	82.63±0.32	81.66±1.51	81.96±0.35	81.63±2.58	86.93±1.47	80.29±0.19	80.30±0.22	88.44±0.70	**88.57±0.83**
AA	82.26	83.63	85.09±0.32	84.73±1.48	83.16±0.95	85.92±2.29	88.42±1.48	82.95±0.44	81.99±0.19	**90.10±0.70**	89.92±0.71
Kappa	77.88	79.14	81.17±0.35	80.11±1.65	80.50±0.41	80.12±2.72	85.82±1.60	78.74±0.18	78.8±0.24	87.48±0.73	**87.62±0.86**

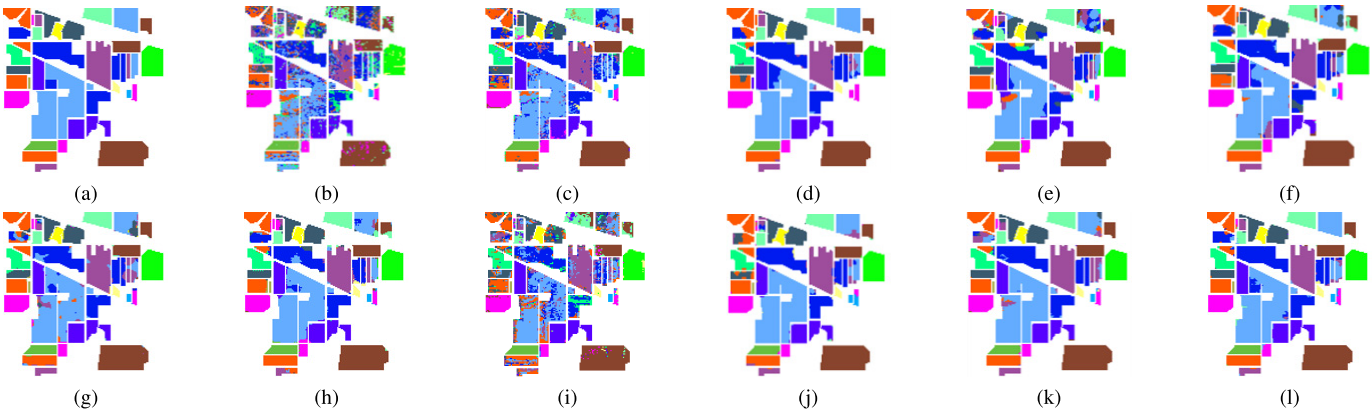

Fig. 8. Classification maps of different methods on the IP dataset. (a) Ground truth map. (b) SVM (OA = 64.88%). (c) EMP-SVM (OA = 82.47%). (d) SGL (OA = 94.70%). (e) HybridSN (OA = 85.20%). (f) pResNet (OA = 88.05%). (g) A^2S^2K-ResNet (OA = 92.77%). (h) MSRN (OA = 91.91%). (i) FuNet (OA = 73.46%). (j) MDGCN (OA = 93.23%). (k) CEGCN (OA = 94.16%). (l) EMS-GCN (OA = 95.87%).

D. Discussion

1) Ablation Study: Here, we carry out a set of ablation experiments to explore how each core module of EMS-GCN contributes to the performance gain. The overall accuracies with the combination of different modules on the three HSI datasets are summarized in Table IX, where the bold number denotes the highest value and the underline number represents the second highest value. The baseline method here denotes the model that applies a superpixel segmentation preprocessing on the raw HSI instead of applying the differentiable superpixel segmentation module, uses the vanilla graph convolutional layer rather than mixhop graph convolutional layer and removes the channel attention module. Note that the multiscale CNN in the differentiable superpixel segmentation module is retained for the baseline model to ensure fairness. Moreover, we adopt "Differentiable Superpixel Segmentation w/o β" to represent the differentiable superpixel segmentation module without learnable parameter β, and "Mixhop Graph Convolutional Layer w/o Update" to represent the mixhop graph convolutional layer without the superpixel graph structure update submodule.

As shown in Table IX, the differentiable superpixel segmentation module without the learnable parameter can bring an increase of 0.44%, 0.96%, and 0.47% in accuracy on the IP, PU, and HU datasets, respectively, mainly because this module makes it possible to produce and optimize superpixels on the basis of the discriminative spectral–spatial features extracted by a trainable multiscale CNN instead of the raw HSI. The accuracy further in creases by introducing the trainable parameter for the calculation of pixel-superpixel

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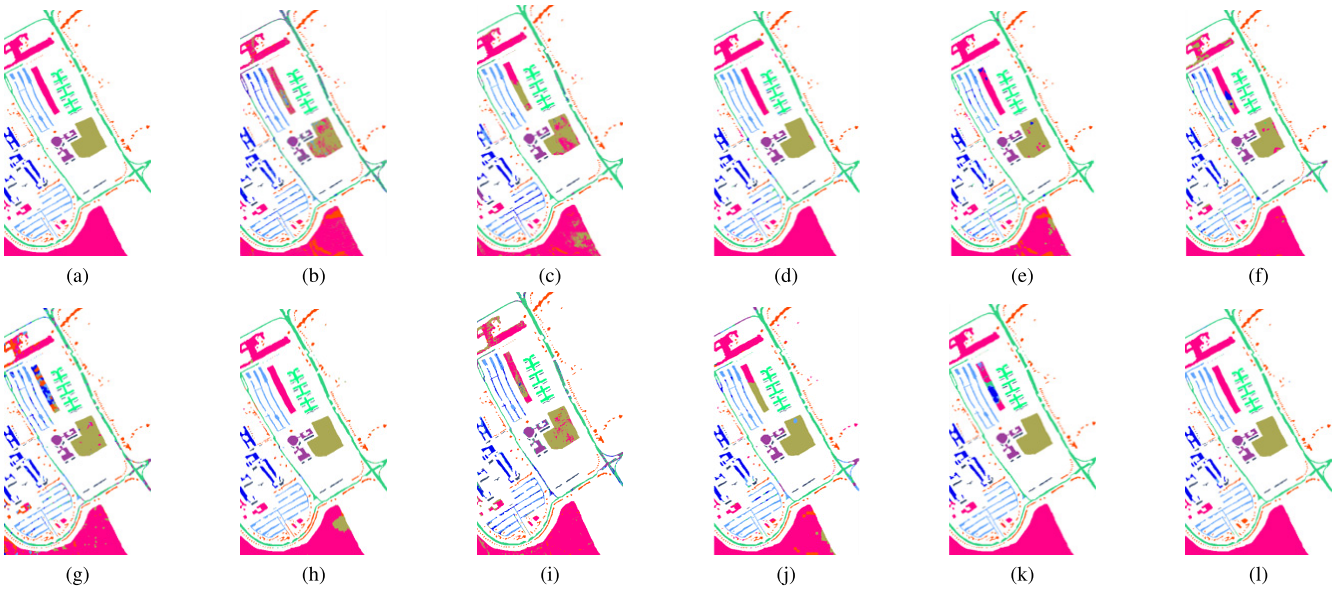

Fig. 9. Classification maps of different methods on the PU dataset. (a) Ground truth map. (b) SVM (OA = 79.72%). (c) EMP-SVM (OA = 87.11%). (d) SGL (OA = 96.69%). (e) HybridSN (OA = 91.34%). (f) pResNet (OA = 93.88%). (g) A^2S^2K-ResNet (OA = 88.44%). (h) MSRN (OA = 96.69%). (j) FuNet (OA = 87.99%). (j) MDGCN (OA = 95.68%). (k) CEGCN (OA = 97.84%). (l) EMS-GCN (OA = 98.47%).

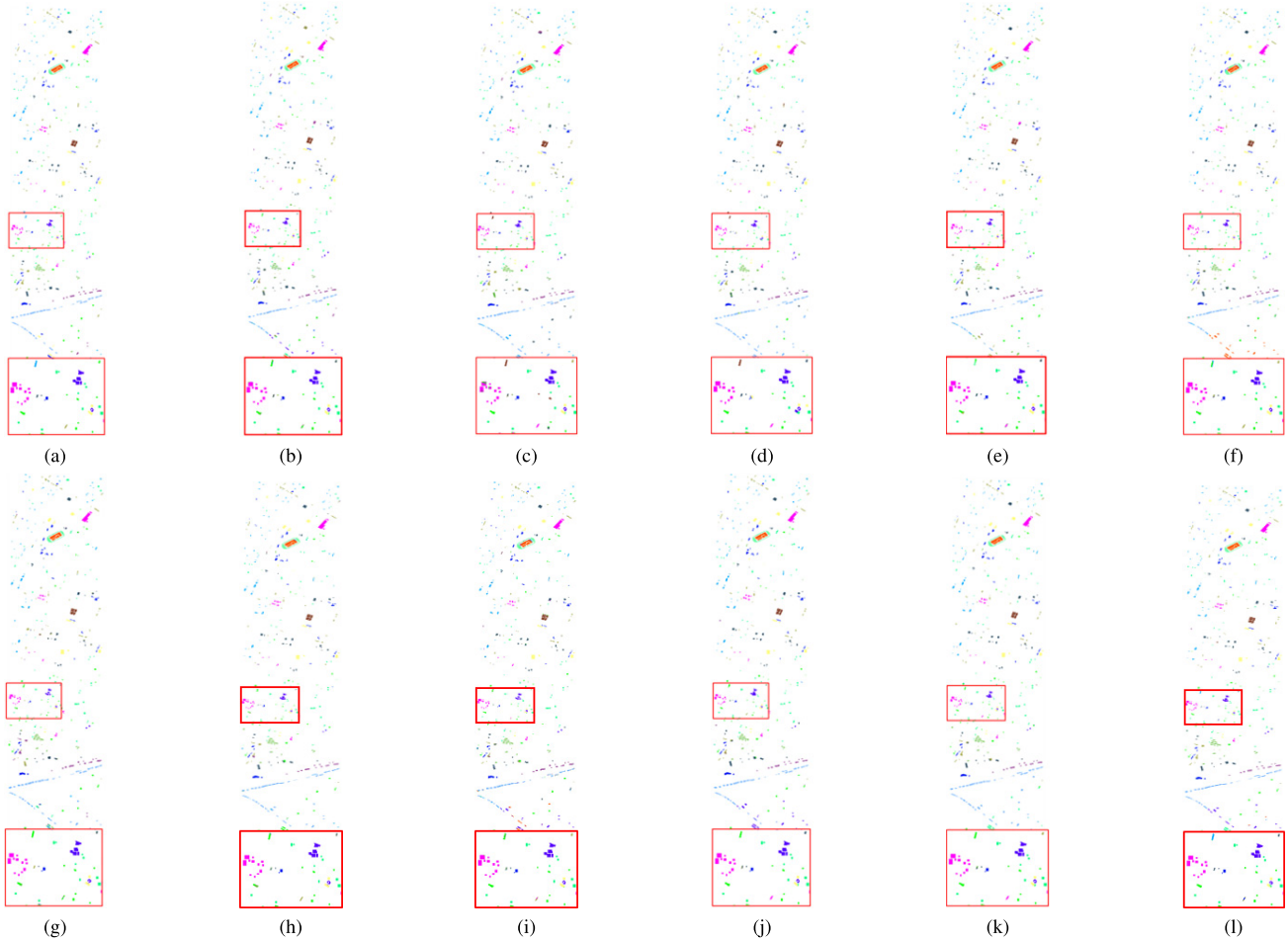

Fig. 10. Classification maps of different methods on the HU dataset. (a) Ground truth map. (b) SVM (OA = 79.51%). (c) EMP-SVM (OA = 80.77%). (d) SGL (OA = 82.63%). (e) HybridSN (OA = 81.66%). (f) pResNet (OA = 81.96%). (g) A^2S^2K-ResNet (OA = 81.63%). (h) MSRN (OA = 86.93%). (i) FuNet (OA = 80.29%). (j) MDGCN (OA = 80.30%). (k) CEGCN (OA = 88.44%). (l) EMS-GCN (OA = 88.57%).

association matrix, which helps to learn an adaptive and flexible association matrix for different datasets. In addition, with the help of the mixhop graph convolutional layer without structure update, the model increases accuracy significantly

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TABLE IX
OA (%) INDICES OF THE COMBINATIONS OF DIFFERENT MODULES ON THREE DATASETS

Baseline	Differentiable Superpixel Segmentation w/o β	Differentiable Superpixel Segmentation	Mixhop Graph Convolutional Layer w/o Update	Mixhop Graph Convolutional Layer	Channel Attention Module	OA		
						Indian Pines	Pavia University	Houston University
✓	-	-	-	-	-	94.21±0.95	92.75±2.01	82.04±1.01
✓	✓	-	-	-	-	94.65±0.43	93.71±1.78	82.51±1.61
✓	-	✓	-	-	-	95.08±1.12	94.23±2.22	83.06±0.76
✓	-	✓	✓	-	-	95.33±0.35	96.92±1.84	87.70±0.20
✓	-	✓	-	✓	-	95.75±0.85	97.77±0.79	88.20±0.59
✓	-	✓	-	✓	✓	**95.87±0.74**	**98.47±0.73**	**88.57±0.83**

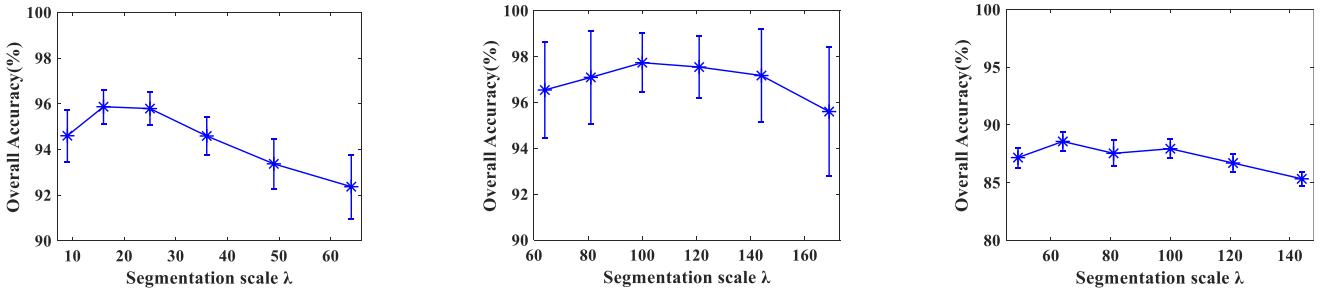

Fig. 11. OA (%) indices of EMS-GCN with different segmentation scales on each dataset. (a) IP. (b) University of Pavia. (c) HU.

by 2.69% of OA on the PU dataset and by 4.64% of OA on the HU dataset, since the local information ignored by the vanilla graph convolutional layer is especially useful for the datasets that include many objects with small-scale and irregular shape. Moreover, it is observed that the superpixel graph structure update module leads to higher performance, which demonstrates that the weighted adjacency matrix can better represent the complex relationship between adjacent nodes than the binary adjacency matrix. As expected, by assigning different weights for the local and long-range information adaptively, the channel attention module can further improve the performance on all datasets to some extent. In a word, with the modules mentioned above, the classification and generalization ability of the model can be significantly enhanced.

2) Parameter Sensitivity Analysis: The EMS-GCN method consists of two hyperparameters that derive from two modules.

1) *Differentiable Superpixel Segmentation:* λ (superpixel segmentation scale).
2) *Mixhop Superpixel-Based GCN:* D (depth of GCN).

In this section, the parameter sensitivity of EMS-GCN regarding to different values of these two hyperparameters is investigated. Specifically, in order to explore how the superpixel number influences the accuracy, we analyze the classification accuracy when the segmentation scale λ is in the range of {64, 81, 100, 121, 144, 169} for the PU dataset, {49, 64, 81, 100, 121, 144} for the HU dataset, and {9, 16, 25, 36, 49, 64} for the IP dataset since it is relatively small. The mean and standard deviation of OA indices on the three datasets are shown in Fig. 11. According to Fig. 11, the classification accuracy of the EMS-GCN is relatively stable with the change of segmentation scale λ, which suggests the robustness of the EMS-GCN method. And EMS-GCN yields the best performance when λ is set to 16 on the IP dataset, 100 on the PU dataset, and 64 on the HU dataset. A smaller λ means more superpixel numbers and less spatial contextual information contained in superpixels. In contrast, a larger λ indicates fewer superpixel numbers, which will lead to "mixed superpixels" that include pixels of different classes in individual superpixel.

For another parameter D, we set it in the range of {1, 2, 3, 4, 5} and test the classification accuracy on each dataset. The related experimental results on the three datasets are provided in Fig. 12. It has been verified that the performance of Graph Neural Network (GNN) models usually degrades when using more than two layers due to the over-smoothing effect. However, as shown in Fig. 12, with the aid of local detailed information extracted by the pixel-based CNN branch or mixhop graph convolutional layer, the CEGCN and our proposed EMS-GCN methods can mitigate the over-smoothing problem to a degree. Furthermore, rather than treating local and long-range information equally in CEGCN, the channel attention module in EMS-GCN learns different weights for the local and long-range information adaptively, which can further improve the classification performance as D increases.

3) Impact of the Number of Training Samples: Obtaining labels in real-world scenarios is notably difficult, hence the classification models are supposed to be equipped with small sample learning capacity. In this section, the impact of the number of training data on comparative methods is examined. For all three datasets, different numbers of pixels (from 5 to 30 pixels at an increment step of 5 pixels for each class) are randomly chosen as the training set. The OA indices of comparative methods with different numbers of

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TABLE X
COMPARISONS IN TERMS OF THE TRAINABLE PARAMETERS, FLOPS, TRAINING TIME, AND TEST TIME OF DIFFERENT DL-BASED METHODS

Dataset	Complexity/Time	HybridSN	pResNet	A^2S^2K-ResNet	MSRN	FuNet	MDGCN	CEGCN	EMS-GCN
Pavia University	Params (K)	533.75	1110.17	220.73	138.44	105.05	**6.81**	152.86	201.69
	FLOPs (M)	16.06	28.43	130.79	40.10	**4.16**	17834.06	13094.82	15510.76
Indian Pines	Train. (s)	**13.95**	278.99	372.06	28.70	83.95	39.04	30.90	23.02
	Test. (s)	1.50	1.16	13.88	1.68	3.68	3.05	0.75	**0.02**
Pavia University	Train. (s)	42.32	547.05	450.40	**21.29**	32.38	159.93	197.03	71.01
	Test. (s)	2.78	2.45	70.99	5.98	38.21	15.04	6.26	**0.14**
Houston University	Train. (s)	**21.44**	593.49	2393.08	98.74	443.41	256.96	172.93	293.11
	Test. (s)	0.95	1.47	20.75	1.42	3.12	58.67	14.28	**0.62**

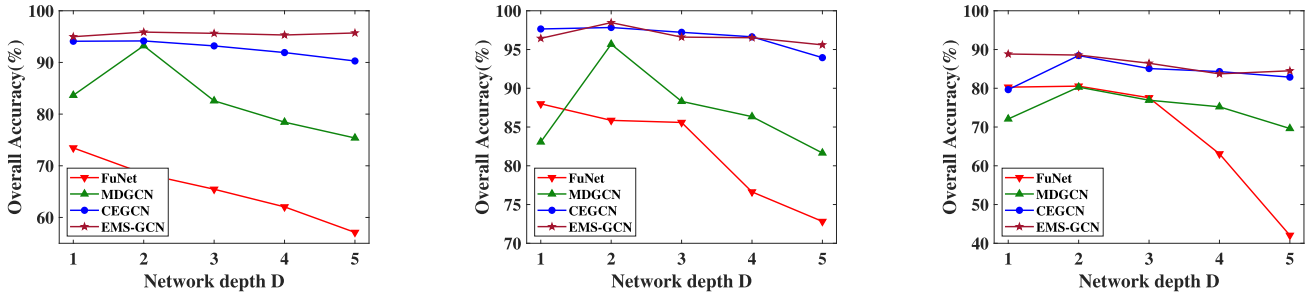

Fig. 12. OA (%) indices of EMS-GCN with different network depth on each dataset. (a) IP. (b) University of Pavia. (c) HU.

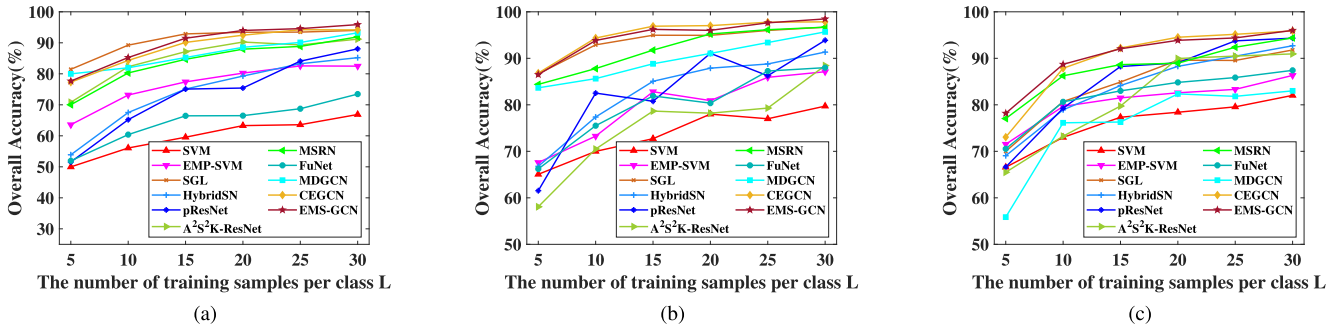

Fig. 13. OA (%) indices of EMS-GCN with different number of training samples on each dataset. (a) IP. (b) PU. (c) HU.

training samples are reported in Fig. 13. Not surprisingly, the performance of all methods improves with the increase of the training sample size. It is especially obvious for the CNN-based methods which require large amounts of training samples. Moreover, EMS-GCN consistently yields better or comparative performance over the other comparative methods on all three datasets, which indicates the outstanding small sample classification capacity and robustness of our method.

4) Visualization Analysis: To further confirm the effectiveness of the key components in EMS-GCN, we first visualize the segmentation maps and the classification results obtained by the SLIC algorithm and the differentiable superpixel segmentation module on the PU dataset. In Fig. 14, the first three pictures present the segmentation map, classification map, and classification difference map acquired by the SLIC algorithm. The last three pictures display the corresponding results adopting the differentiable superpixel segmentation module. According to Fig. 14, the differentiable superpixel segmentation module helps to reduce the misclassification in the boundary regions by generating precise superpixel boundaries that closely match the irregular objects boundaries. In addition, the feature maps obtained by the vanilla GCN and mixhop GCN on the PU dataset are displayed. Note that the channel attention module is added to the vanilla GCN for fairness. In Fig. 15, the first two pictures show the feature maps in the first and second layers of vanilla GCN. The last two pictures display the corresponding feature maps of mixhop GCN. It is manifest that feature maps of vanilla GCN are indistinct and suffer from edge details loss. In contrast, with the benefit of the mixhop graph convolutional layer, more detailed information and sharper edges can be preserved in the feature map.

To analyze the class separation capability of features extracted by different methods, we apply the t-Stochastic Neighbor Embedding (SNE) technique to visualize the high-dimensional outputs of different deep networks on the PU dataset in Fig. 16. Generally, samples from different classes are distributed more distinctly and compactly in the T-SNE visualization of the proposed EMS-GCN method compared with

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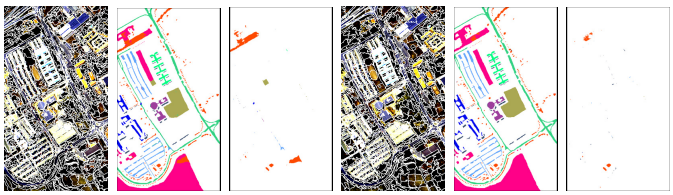
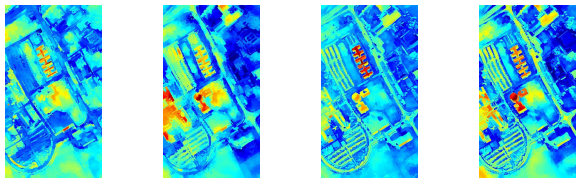


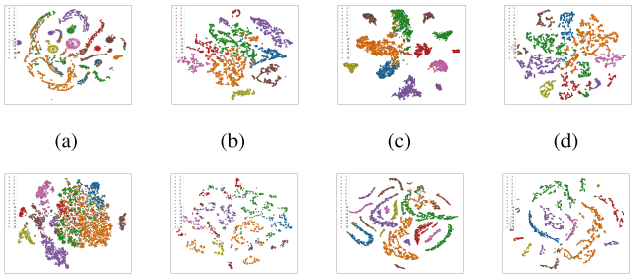
(a) (b) (c) (d) (e) (f)

Fig. 14. Demonstration on the effect of the differentiable superpixel segmentation module on the PU dataset. (a) Superpixel segmentation map of SLIC algorithm. (b) Classification map of SLIC algorithm of SLIC algorithm. (c) Classification difference map of SLIC algorithm. (d) Superpixel segmentation map of differentiable superpixel segmentation module. (e) Classification map of SLIC algorithm of differentiable superpixel segmentation module. (f) Classification difference map of differentiable superpixel segmentation module.



(a) (b) (c) (d)

Fig. 15. Demonstration on the effect of the mixhop graph convolutional layer on the PU dataset. (a) Feature map in the first vanilla graph convolutional layer. (b) Feature map in the second vanilla graph convolutional layer. (c) Feature map in the first mixhop graph convolutional layer. (d) Feature map in the second mixhop graph convolutional layer.



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 16. T-SNE feature visualization of different comparison methods on the PU dataset. (a) HybridSN. (b) pResNet. (c) A2S2K-ResNet. (d) MSRN. (e) FuNet. (f) MDGCN. (g) CEGCN. (h) EMS-GCN.

other methods. For example, it is observed that the samples from categories 2 and 11 are seriously mixed for the T-SNE visualization of other methods. Unlike the other comparative methods, our proposed EMS-GCN method can distinguish the two categories to a great degree. As a result, the EMS-GCN method can significantly maximize the interclass variance and minimize the intraclass variance and yield better classification performance both qualitatively and quantitatively.

5) Model Complexity and Efficiency Analysis: The number of trainable parameters and the number of arithmetic operations [also Floating Point of Operations (FLOPs)] are adopted to measure the model complexity of different DL-based methods on the PU dataset. Meanwhile, we use the training and test time on three datasets to evaluate the computational efficiency. Note that the time of superpixel segmentation is included in the running time for GCN-based methods. As reported in Table X, the superpixel-based GCN methods require fewer trainable parameters compared with other methods, while their FLOPs are hundreds of times higher than CNN-based methods. This can be explained by that FLOPs are calculated at batch size 1, and the superpixel-based GCN methods directly fed the whole image instead of a small patch into the GCN models for each batch. When considering all patches (i.e., 207 400 patches for the PU dataset) for other models, the superpixel-based GCN-based methods still require less computation. In addition, the training time of the CNN-based methods is always longer than GCN-based methods because of the redundant computation. However, HybridSN and MSRN are the exception due to their fast convergence speed. In terms of the test time, the proposed EMS-GCN method provides the fastest inference speed on three datasets, (i.e., 0.02, 0.12, and 0.64 s for testing over the IP, PU, and HU datasets) owing to two optimizations of the most time-consuming differentiable superpixel segmentation module: 1) The size of the assignment matrix in the differentiable superpixel segmentation module is reduced from $n \times m$ to $n \times 9$ by constraining the distance computations as suggested in [44]. 2) The differentiable superpixel segmentation module can be conducted in GPU, while the traditional superpixel segmentation techniques are Central Processing Unit (CPU)-only algorithms. Therefore, the proposed method is qualified for practical applications.

IV. Conclusion

In this article, we have proposed an EMS-GCN method for HSI classification, which combines the modified CNN-based superpixel segmentation method with the mixhop superpixel-based GCN to optimize superpixel boundary as well as adaptively extract both the local and long-range information. First, the differentiable superpixel segmentation module is introduced to generate superpixel representations, which allows refining superpixels with the training of the network. Then, the superpixel graph is constructed with the superpixel representations and binary adjacency matrix. After that, we devise a novel mixhop GCN to adaptively aggregate the local and long-range features of the superpixel graph as well as update the graph structure. Finally, the enhanced superpixel representations are mapped back into pixel feature representations to conduct pixel-wise classification. Extensive experiments on three widely used HSI datasets indicate the effectiveness and robustness of our proposed EMS-GCN method.

Nevertheless, there is still incredible room for improving the latent graph structure of HSI data. In the future, how to exploit the graph structure learning techniques for automatic graph construction will be a desirable direction. Moreover, we will investigate the graph pooling operation to learn hierarchical representations of HSI graph.

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